

Course: Topics in Graph Algorithms

Course Code:

Course Description: Spectral techniques for graphs mainly refer to the study of eigenvalues and eigenvectors of the matrices associated with those graphs. These techniques are becoming increasingly important in designing graph-theoretic algorithms especially while dealing with large volume of data. This massive volume of data do not permit use of algorithms which do not scale linearly with the size of the data. Hence we require faster algorithms for many graph problems. Often many of these problems are solved using graph Laplacian. And one of the showstoppers in this area of spectral graph methods is the development of linear time algorithm for Laplacian solver by Spielman and Teng. As a consequence we got faster algorithms for many important graph problems. This course aims to study such algorithms which are enriched by the interplay between linear algebraic and probabilistic tools in one hand and graph theoretic tools on the other.

We make an entry into this exciting field by exploring few important relationships between eigenvalues, eigenvectors of adjacency matrices, Laplacian matrices of graphs with their structural parameters. We shall also design few algorithms using spectral parameters of graphs. They include algorithms for graph-partitioning, maxcut, small set expansion, max-flow. Random walk on graphs which is another technique heavily used in algorithm design uses spectral properties of graphs in very important way. We shall explore this direction too. Random walk in graphs when modeled as electrical network reveal certain interesting links that help to get an idea about parameters like commute time quite easily. We also explore its remarkable connection with graph sparsifiers. In addition, random walk in random graphs are also touched upon to get a glimpse of how algorithms on social networks modeled using random graphs are often analyzed. Finally we aim to study fast algorithms for Laplacian solvers which heavily rely on spectral graph theory.

Prerequisite(s):

Probability – the level of DA104

Linear Algebra – the level of DA109

Note(s): Syllabus changes yearly and may be modified during the term itself, depending on the circumstances. However, students will be evaluated only on the basis of topics covered in the course.

Credit Hours: 4

Text(s):

Spectral and Algebraic Graph Theory

Daniel A. Spielman

Lecture Notes on Expansion, Sparsest Cut, and Spectral Graph Theory

Luca Trevisan

Random Walk on Graphs: A Survey

L. Lovász

Random Graph Dynamics

Rick Durrett

Random Walks and Electric Networks

Peter G. Doyle and J. Laurie Snell

Lecture Notes:

Spectral Graph Theory in Computer Science

Salil Vadhan

<https://salil.seas.harvard.edu/classes/spectral-graph-theory>

Spectral Algorithms

Lap-Chi Lau

<https://appsrv.cse.cuhk.edu.hk/~chi/csc5160/>

Lecture Notes on Spectral Graph Methods

Michael W. Mahoney

<http://arxiv.org/abs/1608.04845v1>

Course Outline (tentative) and Syllabus:

The following are the topics that we intend to discuss. The topics will not be necessarily covered in the order in which they are listed.

1. Linear Algebra Review

- Eigenvalues, Eigenvectors
- Orthogonal Matrices
- Real Symmetric Matrices

2. Graph Spectra and some applications

- Spectra of: Path graphs, Complete graphs, Cycle graphs
- Rayleigh quotient, Courant-Fischer Theorem, Eigenvalues Interlacing Theorem
- Graph connectivity, regular graphs, bipartiteness.
- Wilf's theorem for Graph Coloring
- Graph Laplacian

3. Cheeger's Inequality and Graph Partitioning Problems

- Cheeger's Inequality
- Sparsest Cut
- Planar Graph Drawing
- Higher Order Cheeger Inequalities
- Max-Cut Approximation using last eigenvalue
- Maximum Independent Set– Hoffman's Bound
- small-set expansion
- multi-way spectral partitioning

4. The Power method

- Approximation of largest eigenvalue
- Approximation second largest eigenvalues
- Second smallest eigenvalue of Laplacian

5. Random Walk on undirected Graphs and its applications

- Second eigenvalues and Convergence of random walk
- Sparse-cut in graphs using random walk
- Markov-Chain Monte Carlo - Sampling spanning trees

6. Modeling graphs as Electrical Network

- Electrical flow, electrical resistance
- Random walk, hitting time, cover time
- Spectral sparsification of graphs

7. Random Walk in Random Graphs

- RW on Preferential Attachment Graph
- Voter Model
- RW on semi-random graph

8. Preconditioning

- Conjugate gradient method
- Graph approximation
- Combinatorial Preconditioning by low stretch spanning trees

9. Fast Laplacian Solvers for Graph Problems

10. Maximum flow

- Multiplicative weight update method
- Fast algorithms approximating maximum flows in graphs